

MMS Convergence Rates

June 2026

This document summarizes the current verification results for the incompressible Navier–Stokes solver using the Method of Manufactured Solutions (MMS). A smooth analytical solution was prescribed for velocity and pressure, and the corresponding forcing term was derived and applied to the governing equations.

The tests were performed using Taylor–Hood finite elements of orders Q_2/Q_1 , Q_3/Q_2 , and Q_4/Q_3 . For the Taylor–Hood pair (Q_{k+1}, Q_k) , the expected asymptotic convergence rates are

$$\begin{aligned}\|u - u_h\|_{L^2} &= \mathcal{O}(h^{k+2}), \\ \|u - u_h\|_{H^1} &= \mathcal{O}(h^{k+1}), \\ \|p - p_h\|_{L^2} &= \mathcal{O}(h^{k+1}),\end{aligned}$$

where u, p are manufactured velocity and pressure solutions, u_h, p_h are computed solutions on mesh of cell size h .

Convergence rates were obtained from linear regression of the log–log error plots. The measured rates are compared with these theoretical expectations in Table 1.

Table 1: Measured and theoretical convergence rates for the Taylor–Hood (Q_{k+1}, Q_k) discretization. Reported uncertainties are the standard errors of the fitted slopes.

Degree k	Norm	Measured rate	Expected rate
1	$\ u - u_h\ _{L^2}$	2.93 ± 0.12	3
	$\ p - p_h\ _{L^2}$	2.48 ± 0.06	2
	$ u - u_h _{H^1}$	1.66 ± 0.13	2
2	$\ u - u_h\ _{L^2}$	4.55 ± 0.08	4
	$\ p - p_h\ _{L^2}$	3.11 ± 0.07	3
	$ u - u_h _{H^1}$	3.32 ± 0.04	3
3	$\ u - u_h\ _{L^2}$	5.33 ± 0.14	5
	$\ p - p_h\ _{L^2}$	4.042 ± 0.018	4
	$ u - u_h _{H^1}$	4.18 ± 0.16	4

Table 2: The table 1 rearranged

Degree k	$\ u - u_h\ _{L^2}$		$\ p - p_h\ _{L^2}$		$ u - u_h _{H^1}$	
	Measured	Expected	Measured	Expected	Measured	Expected
1	2.93 ± 0.12	3	2.48 ± 0.06	2	1.66 ± 0.13	2
2	4.55 ± 0.08	4	3.11 ± 0.07	3	3.32 ± 0.04	3
3	5.33 ± 0.14	5	4.042 ± 0.018	4	4.18 ± 0.16	4

The log–log plots of the error norms as a function of cell size are shown below for the three Taylor–Hood discretizations considered. The fitted slopes correspond to the convergence rates reported in Table 1.

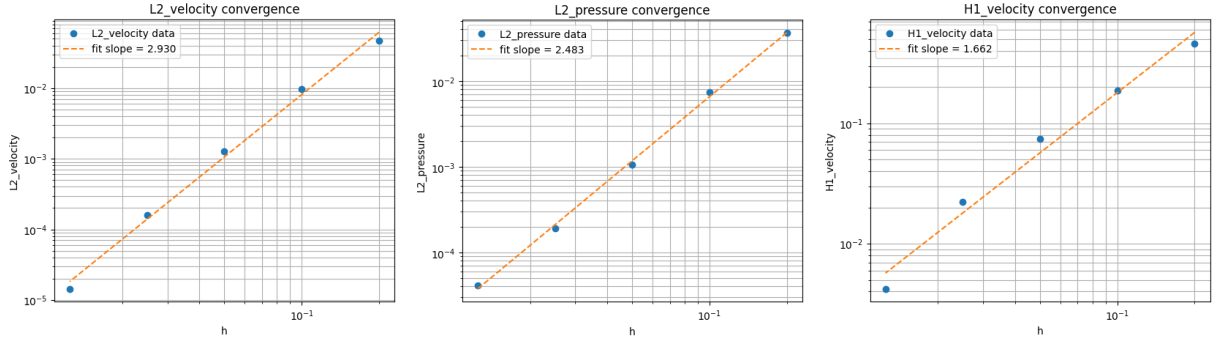


Figure 1: Q_2/Q_1 Taylor-Hood element (degree $k = 1$). Left: velocity L^2 error. Center: pressure L^2 error. Right: velocity H^1 error.

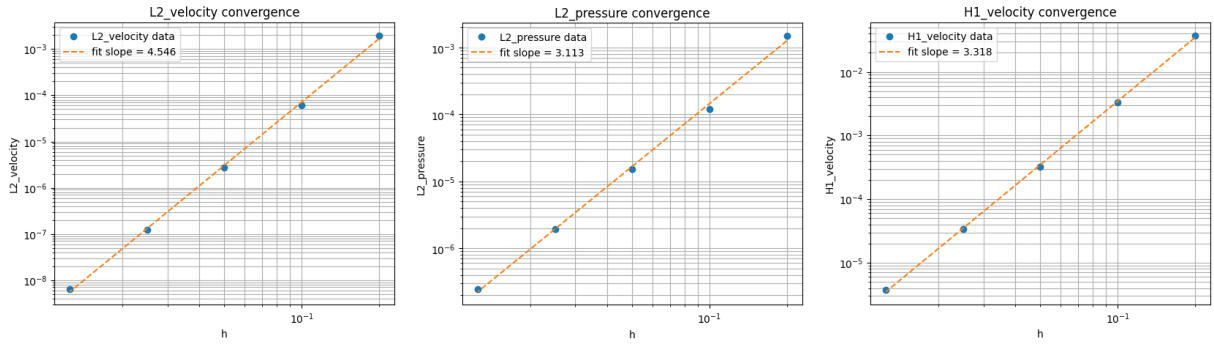


Figure 2: Q_3/Q_2 Taylor-Hood element (degree $k = 2$). Left: velocity L^2 error. Center: pressure L^2 error. Right: velocity H^1 error.

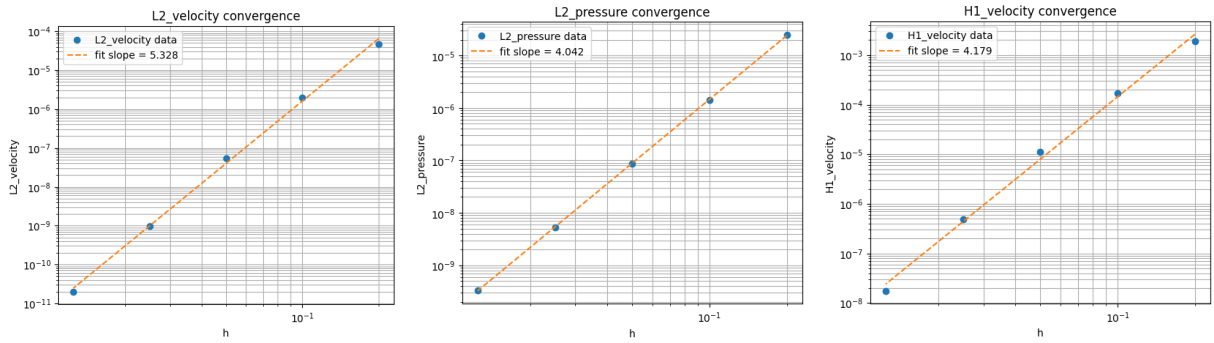


Figure 3: Q_4/Q_3 Taylor-Hood element (degree $k = 3$). Left: velocity L^2 error. Center: pressure L^2 error. Right: velocity H^1 error.